LECTURE 06 | START HERE

Evaluation, usability testing and field trials

A lecture-focused guide with worked calculations, study-design examples and concise recall practice.

Scope and naming

This covers the whole supplied file, including its second half on **data analysis**. "Lecture 06" follows the filename; the timetable places this topic on Oct. 28. These are study notes, not an official assessment outline.

Pages	What to study
2-3	Usability-test procedure, measures and SUS scoring
4-6	Experimental variables/designs, counterbalancing, field studies
7-9	Data types/scales, statistics, boxplots and questionnaire results
10-11	Content analysis, critical incidents and grounded theory
12	Study blueprint, reporting and final recall checklist
13-15	24 short-answer prompts with two-line models
16-20	40 original MCQs, with answers kept separate
21-22	MCQ explanations and source-page references

The main distinctions

Formative / summative: improve a design during development / assess a finished product.
Testing / experiment / field: find product problems / compare controlled conditions / understand real contextual use.
Performance / attitude: observed success, time and errors / users' reported perceptions.
Quantitative / qualitative: numerical measurements / words, images and interpretations.

Use the guide actively

Explain the distinction before looking at options. For a short answer, give the concept in line one and its consequence, example or method in line two. For calculations, show the conversion or numerator/denominator before the result.

Source references use **PDF page numbers**, which sometimes differ from the printed slide footer. Notes distinguish lecture examples from additional explanations and do not claim new study findings.

EVALUATION | TEST DESIGN

Plan a usability test around the goal

Evaluation belongs throughout design

Before design, assess needs and viability. During design, evaluate prototypes and revise them. After design, assess acceptance and use. **Formative** and **summative** describe purpose and timing; "lab" and "field" describe setting. They are not interchangeable categories.

A defensible procedure

1. Define the intended users, tasks and questions.
2. Recruit representative participants; obtain informed consent.
3. Give realistic task goals and define success, timing and assistance rules.
4. Observe task performance; record agreed measures and relevant comments.
5. Gather post-task perceptions using questions or a suitable questionnaire.
6. Analyze problems and consequences; revise and re-evaluate.

The lecture gives **5-12 users** as a typical usability-test range. It is not a universal sample-size guarantee, especially for statistical comparisons or varied user groups.

Question	Candidate measure	What to avoid
Can users achieve the goal?	Correct completion and errors: effectiveness/safety.	Calling speed a substitute for successful completion.
How much effort/time?	Completion time: efficiency.	Comparing times for unequal tasks or hiding failed attempts.
Can new users discover it?	Help requests; time to find the correct path: learnability.	Teaching each action before testing first use.
Can they return after a gap?	Performance after a specified absence: memorability.	Inferring long-term recall from immediate repetition.
How do they experience it?	Comments, hesitation and post-task ratings.	Assuming a smile proves success or a rating explains every error.

Guidance and recording change the evidence

Avoid coaching if learnability is the target. A complex system may need standardized training to reach the part being studied; report it. Give a **goal**, such as finding and enrolling in a course, rather than a menu-by-menu recipe.

The slides explicitly require **written, signed consent before face/voice recordings**. Logs can capture clicks and timing; video can preserve interface action and reactions. Comments explain context; post-task responses capture attitudes and priorities.

MEASUREMENT | SUS

Calculate SUS; interpret it carefully

The **System Usability Scale (SUS)** has ten Likert-style items, usually answered from 1 (strongly disagree) to 5 (strongly agree). It measures perceived usability after interaction; it is not a direct performance test.

Standard alternating form	Conversion	Example
Positive items 1, 3, 5, 7, 9	Contribution = response - 1	Response 4 -> 3 points.
Negative items 2, 4, 6, 8, 10	Contribution = 5 - response	Response 2 -> 3 points.
Overall score	Sum the ten contributions, then multiply by 2.5.	Converted total 30 -> SUS 75.

Worked calculation

Suppose all positive-item responses are 4 and all negative-item responses are 2. Each converted item contributes 3.
Total = 5(4 - 1) + 5(5 - 2) = 15 + 15 = 30.
SUS = 30 x 2.5 = 75. Each item contributes 0-4, so totals span 0-40 and scaled scores span 0-100.

A score is not a percentage

SUS 75 does **not** mean 75% of tasks succeeded or that the score is the 75th percentile. Completion needs performance data; percentile interpretation needs a reference distribution.

Using the lecture's reference table

The lecture gives a reference mean of **68**. Its curved table places **74.1-77.1 in grade B, percentile band 70-79**; therefore the example score 75 falls there. These are reference-table interpretations, not a course grade or a universal pass/fail threshold.

Version and numbering traps

The lecture also mentions an all-positive version. Score according to **actual item wording and the chosen instrument**: the odd/even shortcut applies to the standard alternating form. An all-positive form does not acquire negative items simply because their item number is even.

The slide on PDF p. 20 repeats item number 8: the confidence statement is the ninth item in the standard sequence. Use the ten-item order and polarity, not that numbering typo.

Two-line model: explain a high SUS score with frequent task failures.

SUS reflects reported usability, while task failures are observed performance evidence.
Report both and investigate the mismatch rather than treating the score as proof of effectiveness.

EXPERIMENTS | CAUSAL COMPARISON

Separate interface effects from alternatives

Term	Meaning	Enrolment example
Independent variable (IV)	Factor deliberately varied by the researcher.	Interface condition: old A versus new B.
Dependent variable (DV)	Response measured to assess the effect.	Time to complete the defined enrolment task; errors or success as additional outcomes.
Experimental condition	A particular state/level of the manipulated factor(s).	Perform the specified task using A, or using B.
Nuisance variable	Other variation that affects the outcome and may add noise.	Fatigue, prior experience or interruptions.
Confound	An alternative influence that varies with condition, making attribution ambiguous.	Only A users get slow hardware; only B users get fast hardware.

A concrete study plan

Question: does B reduce successful enrolment time compared with A?
Task: enrol in a specified available course using equivalent starting states.
Conditions: the same task requirements and defined training, using A or B.
Record: time under a consistent start/stop rule, successful completion, errors and assistance.
Control: equivalent hardware, instructions, environment and task difficulty; address relevant experience and order.

If A's users have different equipment or prior training, faster performance with B is not clean evidence for the interface alone. Random assignment helps distribute differences, but cannot guarantee identical groups, particularly with small samples.

Testing a claim is more than looking at the lower mean

Use representative recruitment and an appropriate design/sample size. Report variability and failed tasks. A larger sample does not repair biased recruitment or a confounded comparison. A descriptive difference in one sample is not automatically statistically significant or generalizable.

A comparison answer should include

IV + DV + task/conditions + plausible confounds + participant allocation. If using within-subjects, include an order-control plan. Describe what participants actually do, not just the names of the variables.

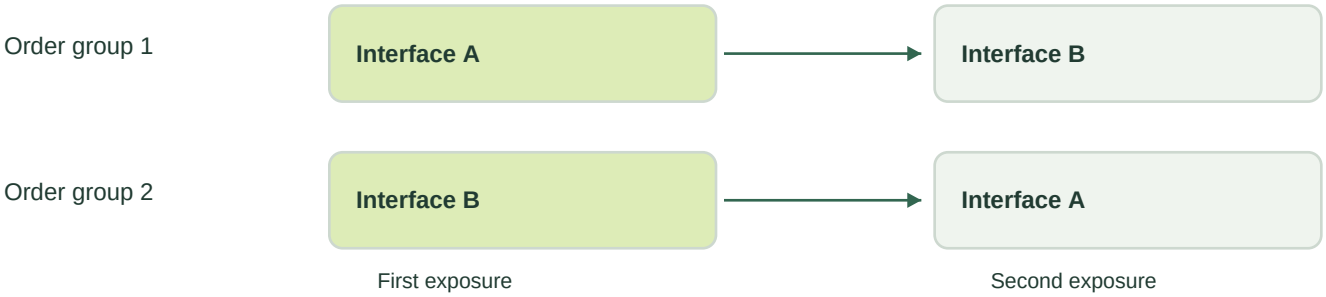
EXPERIMENTS | PARTICIPANT ALLOCATION

Between, within and matched designs

Design	How participants take part	Benefit and limitation
Between-subjects	Separate groups use A or B; randomly allocate where appropriate.	Avoids cross-condition order effects; needs more people and can retain group differences.
Within-subjects	Each participant uses both A and B.	Each person is their own comparison; practice, fatigue and carryover can affect later conditions.
Matched participants	Pair people on relevant characteristics; distribute pair members across conditions.	Reduces selected differences; matching is never perfect on all factors.

Counterbalance order, rather than assuming practice disappears

TWO-CONDITION WITHIN-SUBJECTS COUNTERBALANCING



If everyone does A then B, improvement may reflect learning; fatigue may instead hurt B. AB/BA distributes exposure order. It does not eliminate all possible carryover or replace equal task difficulty, appropriate training and a suitable analysis.

Latin squares for more conditions

A Latin square places each condition once in each sequence position. It uses fewer orders than all possible permutations. A **balanced** Latin square can also balance immediate preceding/following relations; a generic Latin square does not automatically balance every carryover effect.

Illustrative 4-condition balanced orders	Sequence
Group 1	A B D C
Group 2	B C A D
Group 3	C D B A
Group 4	D A C B

These orders reproduce the pattern shown on PDF p. 47. Each condition appears in every position. Distribute participants sensibly across the chosen orders; the square itself does not recruit or randomize them.

STUDY SETTING | FIELD TRIALS

Control and context answer different questions

Approach	Primary question	Typical evidence / limitation
Product usability test	Where does this product obstruct representative tasks, and how should it change?	Planned tasks and performance plus comments; controlled setting may omit real context.
Research experiment	Does a manipulated factor affect an outcome under the specified design?	Controlled comparison, replicability and statistical validation; may simplify natural use.
Field study / field trial	How is technology used, adopted and adapted in its real environment?	Observation, interviews, notes, recordings and logs; reduced control of competing influences.
Analytical evaluation	What problems do inspection or predictive methods identify?	Heuristics, walkthroughs or models; no direct evidence from recruited users in the evaluation itself.

Appropriation: what people actually make of a technology

Users integrate and adapt technology to their needs, desires and culture. A feature used differently from the designers' plan may reveal a useful adaptation or a missing requirement. Field studies can discover opportunities, establish requirements, support introduction and evaluate existing use.

Example: your nutrition companion in daily life

A lab task can reveal whether users can correct a logged meal. A field study can reveal why they delay entry during commuting, share information, ignore reminders in public or record notes outside the app. Those contextual findings can change the requirements.

Use observation and contextual interviews, with agreed logs or recordings where appropriate. Natural use offers realism but does not by itself isolate a causal interface effect. Quantitative logs can supplement predominantly qualitative field evidence.

Triangulate when evidence disagrees

Combine relevant approaches instead of naming one as universally best. If participants give positive ratings but repeatedly abandon a task, examine the performance, comments and context. The disagreement is a finding to investigate.

Two-line model: choose lab or field for a reminder that fails during real work.

A field study reveals how interruptions and workplace conditions affect the reminder in actual use.
Follow with controlled testing if you need to isolate the effect of a specific redesign.

ANALYSIS | DATA TYPES AND SCALES

The recorded variable determines the scale

Collection method is not the same as data type

Quantitative: numerical observations such as time, error counts and closed-response counts. **Qualitative:** words, images, descriptions and recorded experience. An interview may collect both a numerical age and an open explanation. Observation can yield a narrative or a count of errors.

Scale	What is meaningful?	Example / useful summary
Nominal	Category identity; no inherent order. Numeric labels remain labels.	User group or country; counts, percentages and mode. Do not average category codes.
Ordinal	Order, without guaranteed equal gaps.	Preference ranks, age brackets, individual agreement responses; frequencies, median, percentiles.
Interval	Order and equal differences; zero is conventional.	Celsius temperature; differences, mean and standard deviation where appropriate. Ratios of values are not meaningful.
Ratio	Equal intervals plus a true, meaningful zero.	Task duration, length or weight; ratios are meaningful, such as 40 seconds being twice 20 seconds.

Three traps that change the answer

Exact age versus brackets: an exact age is a duration with a meaningful zero; recorded age brackets are ordered categories. Classify the data collected, not the broader topic.
Single Likert item: 1-5 agreement labels are ordered; numbers alone do not prove equal distances or a true zero.
Zero versus negative: interval/ratio is decided by the meaning of zero, not simply whether negative values appear in a dataset.

The lecture's apple-liking example

PDF p. 68 illustrates an evenly marked 0-20 liking line as interval. That example assumes equal spacing in the measurement interpretation; it does not establish that every verbal agreement category has equal psychological gaps. If asked about the slide's example, name its assumption; for a single categorical Likert item, ordinal is the cautious classification.

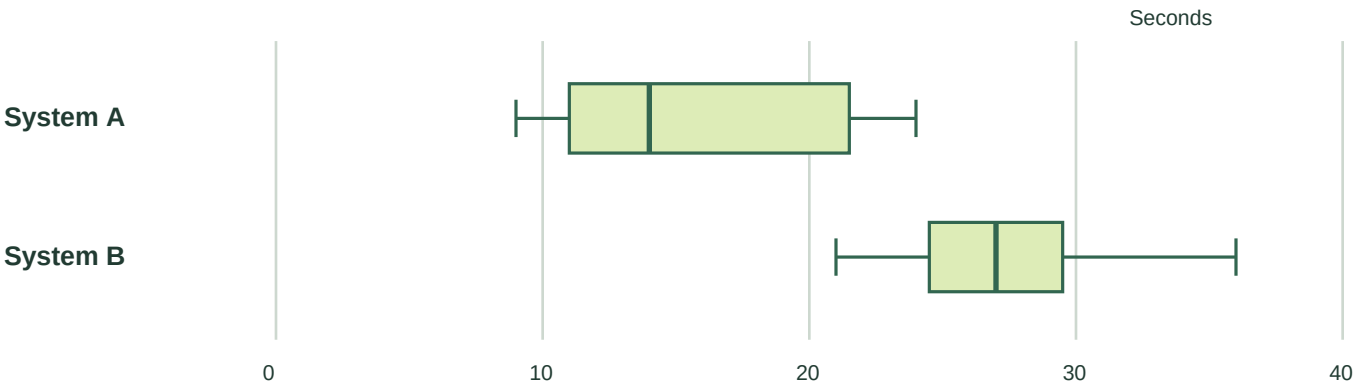
The age-bracket quiz on PDF p. 69 also omits exactly age 20 ("below 20" then "21-30"). A usable questionnaire needs exhaustive as well as non-overlapping categories. Its ordering is still ordinal.

ANALYSIS | DESCRIPTIVE STATISTICS

Summarize typical values and spread

Statistic	Meaning	For 2, 2, 3, 4, 19 seconds
Mean	Sum divided by count. Sensitive to extreme values.	30 / 5 = 6 seconds.
Median	Middle sorted value; average the two middle values for an even count.	3 seconds.
Mode	Most frequent value; ties can produce several modes.	2 seconds.
Range	Maximum minus minimum.	19 - 2 = 17 seconds.
IQR	Third quartile minus first: spread of the middle half.	State the quartile convention if calculating a small-sample IQR.
Standard deviation	Spread of numerical values around their mean.	Not the same as the standard error or a confidence interval.

Read the lecture's nine-person comparison



A: times 9, 12, 12, 14, 16, 19, 24, 10, 24 seconds; median **14**, mean **15.56**.
B: times 21, 25, 24, 28, 30, 36, 26, 27, 29 seconds; median **27**, mean **27.33**.

The box shows Q1-Q3; the interior line is the median. In this illustration whiskers show **minimum and maximum**. Other boxplots use different whisker/outlier rules, so read the caption instead of assuming every whisker is the maximum.

A careful conclusion

These nine people appear faster with A on this task. This descriptive pattern alone does not prove statistical significance, explain the cause, or establish superiority for all users or all usability goals.

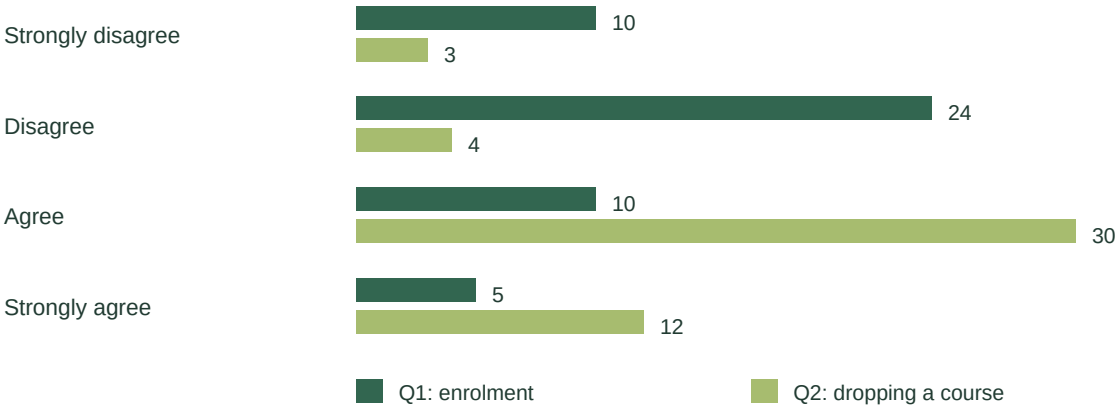
Confidence interval: an uncertainty interval for a parameter/estimate under a specified statistical procedure. A 95% confidence interval is not the range containing 95% of individual users' task times. Formal calculations are not developed in these slides.

ANALYSIS | VISUALIZE AND INTERPRET

Do the arithmetic; limit the claim

Display	Good question	Caution
Bar chart	How do counts/values compare between categories?	Use clear categories, units and a suitable baseline.
Scatterplot	How do two numerical variables relate?	An association does not establish causation.
Boxplot	How do distribution centres/spreads compare?	Show sample size and state whisker rules.
Pie chart	What are the parts of a meaningful whole?	Many similar segments can make comparison difficult.

Lecture example: 49 questionnaire respondents



Q1: enrolment is easy. Positive = 10 agree + 5 strongly agree = 15.
15 / 49 x 100 = 30.6%.
Q2: dropping a course is easy. Positive = 30 agree + 12 strongly agree = 42.
42 / 49 x 100 = 85.7%.

The slide reports approximately 30% and 85%. Keep the original counts and use a consistent rounding rule rather than copying the approximations as exact percentages.

What does this support?

In this sample, dropping a course is rated easier than enrolling. Enrolment may warrant further usability investigation. These are **self-reported judgments**; the table does not establish measured completion times, error rates or a causal explanation for the difference.

Two-line model: interpret the questionnaire comparison.

Positive ratings were 15/49 (30.6%) for enrolment and 42/49 (85.7%) for dropping a course.
Respondents rated enrolment less favourably; observe its tasks to diagnose the underlying problems.

QUALITATIVE ANALYSIS | CODING

Make interpretations traceable to evidence

Content analysis: a systematic route from excerpts to findings

Define a categorization scheme, either before coding or as it emerges from the data. Apply it to meaningful units: sentences, events or audio/video excerpts. Examine patterns and counts, retaining the underlying examples and definitions so the reasoning can be checked.

Evidence from the cashier example	Possible code	What it does / does not show
The cashier describes frustration after scanning fails.	Frustration / experience problem.	Supports a reported experience problem, not a universal claim about every cashier.
The cashier uses manual barcode entry after a failed scan.	Workaround / manual entry.	Shows a process adaptation; investigate its frequency and consequences.
The cashier says manual entry is easy to get wrong.	Reported error risk / safety concern.	Indicates concern, not proof that a particular transaction was actually entered wrongly.

Coding consistency and frequency

Inter-rater reliability asks how consistently different people code the same material. Use clear definitions and examples, compare assignments and investigate disagreement. Agreement improves confidence in consistency; it does not by itself prove that the categories capture the right construct.

The slides' category numbers/labels shift between PDF pp. 75-77. Use stable **named categories** in your own work. Define whether a count means excerpts, events or participants; several comments by one person are not automatically several different users. Frequent does not necessarily mean severe.

Critical incidents concentrate on pivotal facts

Prefer specific events over broad impressions. If a failed scan leads to a wrong manually entered barcode and a billing dispute, analyze that sequence and consequence. "People were unhappy" loses the actionable mechanism. Critical incidents may illustrate either success or failure.

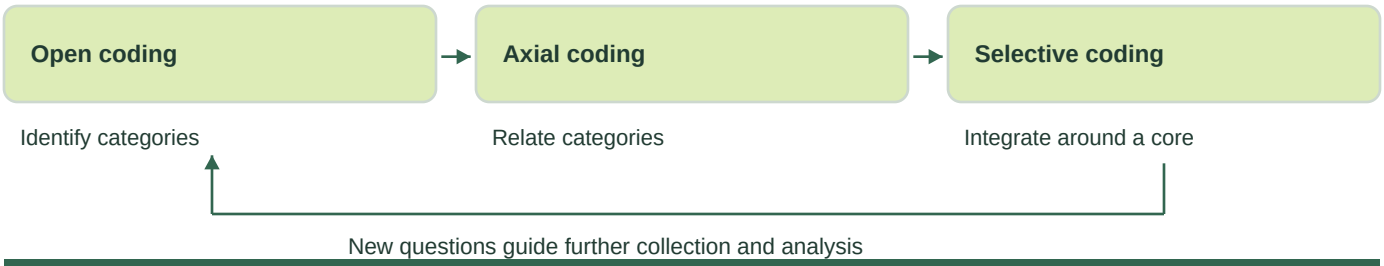
A reportable finding

Observation/excerpt -> code/pattern -> interpretation -> design implication -> limitation. For example, manual-entry errors suggest reviewing input validation and recovery; test the redesign before claiming it solved the problem.

QUALITATIVE ANALYSIS | GROUNDED THEORY

Build the account through repeated analysis

Grounded theory develops a theoretical account from empirical evidence through repeated data collection and analysis. Early categories and relationships guide what to investigate next; the account is refined as new evidence arrives.



Coding stage	What the researcher does	Lecture example
Open coding	Label relevant concepts/events, then develop categories from raw data.	Long commuting; full-time jobs; low attendance; video presentations; eClass.
Axial coding	Relate categories and subcategories; explore those relationships in further collection.	Commuting and employment connect to barriers to physical access; videos may address those barriers.
Selective coding	Organize categories around a central integrating category/account.	Increasing inclusivity and access through video presentations.

What makes it grounded?

The explanation must remain connected to evidence, not merely to a theory chosen in advance. Re-read data, examine exceptions and revise categories/relationships. Do not treat "fits the first dataset" as proof of validity or universal applicability. Grounding is a systematic research process, not a guarantee of a perfect theory.

How this differs from hypothesis testing

A hypothesis-led experiment starts with a claim to test using a planned comparison. Grounded theory emphasizes developing the account through collection/analysis cycles. Both need evidence; naming the method does not make an inference correct.

Activity theory and tools - introductory scope

The lecture briefly contrasts **activity theory** as a pre-existing conceptual framework for interpreting activity. It does not develop its full framework here. Spreadsheets/statistical packages support numerical work; NVivo/Atlas.ti support coding and organization. Software helps manage evidence but does not supply the interpretation for you.

Two-line model: summarize the three coding stages.

Open coding identifies categories; axial coding explores relationships among categories and subcategories.
Selective coding integrates them around a central account, refined through further evidence.

APPLY AND REVIEW

From evaluation question to useful finding

A practical blueprint for your nutrition-app project

Question: can representative users log a meal, correct a mistaken quantity and understand the updated summary?
Purpose: formative usability evaluation of the prototype.
Task: provide a realistic meal-entry goal without revealing the button sequence.
Measure: successful correction, time, errors/help, comments and post-task perceived ease.
Procedure: consent first; consistent instructions; record agreed evidence; preserve task context.
Follow-up: examine actual reminder use in context if natural interruptions are the issue.

Recall this	Minimum useful explanation
Formative / summative	During design to improve / at completion to assess.
Test / field / experiment	Product task problems / natural contextual use / controlled comparison.
SUS	Convert by item polarity; sum ten contributions; multiply by 2.5. Score is not a success percentage.
IV / DV / confound	Manipulated factor / measured response / competing influence tied to condition.
Between / within / matched	Separate groups / same people / pairs matched on relevant characteristics.
Counterbalance	AB and BA; for more conditions, planned balanced orders such as a Latin square.
Scales	Nominal labels; ordinal orders; interval equal gaps; ratio meaningful zero.
Statistics	Centre plus spread; inspect distributions; report sample limits and failures.
Qualitative methods	Systematic content coding; inter-rater agreement; pivotal critical incidents.
Grounded theory	Open -> axial -> selective, with repeated evidence collection and analysis.

Report findings with a reason to act

State the users, tasks, setting and method; show key evidence with units/denominators; explain the problem and consequence; recommend a targeted change; acknowledge limits and plan the next check. A task-time difference, rating or code count is evidence to interpret, not a sales claim.

Source: A. Pavlovych, *3230 06 Evaluation UsabilityTesting-FieldTrials.pdf*, supplied 94-page lecture. This guide summarizes its substantive content and adds original study examples, calculations and practice. It does not reproduce all administrative announcements or claim to replace instructor updates.

ACTIVE RECALL | SHORT ANSWERS

Two-line models 1-8

Recall before reading the answer. Keep both essential points; equivalent accurate wording can count.

1. Distinguish formative from summative evaluation.

Formative evaluation finds problems during design to guide the next iteration.
Summative evaluation assesses the success of a finished product at the end of a design cycle.

Lecture 06 | PDF p. 12

2. Distinguish analytical from empirical evaluation.

Analytical evaluation inspects or predicts interaction without observing recruited users perform tasks.
Empirical evaluation collects evidence from users, through usability tests or field studies.

Lecture 06 | PDF pp. 5, 10-11

3. Why should task instructions avoid teaching every action when testing learnability?

Step-by-step guidance conceals whether users can discover the interaction themselves.
Give a clear task goal and report any help or training so results remain interpretable.

Lecture 06 | PDF pp. 14-15

4. Give one measure each for effectiveness, efficiency and memorability.

Effectiveness: successful task completion; efficiency: time to complete the task.
Memorability: task performance after a specified period away from the system.

Lecture 06 | PDF p. 16

5. What must happen before recording a participant's face and voice?

Explain participation and the intended recording, and obtain informed consent.
The lecture requires written, signed consent before making face or voice recordings.

Lecture 06 | PDF pp. 14, 17

6. Why combine task-performance data with comments and post-task ratings?

Performance data show task success, time and errors; comments can explain the observed difficulties.
Post-task ratings add perceived usability and experience, which may differ from observed performance.

Lecture 06 | PDF pp. 16-18, 22-23

7. Explain how to calculate a standard SUS score.

Odd positive items contribute response minus 1; even negative items contribute 5 minus response.
Sum the ten converted values and multiply by 2.5 for a score from 0 to 100.

Lecture 06 | PDF pp. 19-20

8. All five positive SUS responses are 4 and all five negative responses are 2. Calculate the score.

Each item contributes 3: positive 4 - 1 and negative 5 - 2.
The total is 10 x 3 = 30; SUS = 30 x 2.5 = 75.

Lecture 06 | PDF p. 19

ACTIVE RECALL | SHORT ANSWERS

Two-line models 9-16

Recall before reading the answer. Keep both essential points; equivalent accurate wording can count.

9. Why does SUS 75 not mean 75% of tasks were completed?

SUS is a scaled questionnaire score summarizing perceived usability, not observed task success.
Measure completion separately; use a reference distribution if interpreting SUS percentiles.

Lecture 06 | PDF pp. 19-23

10. Identify the independent and dependent variables in a speed comparison of interfaces A and B.

The independent variable is the interface condition: A or B.
The dependent variable is measured task-completion time for the defined task.

Lecture 06 | PDF pp. 30-32

11. Old-interface users get slow computers and new-interface users get fast ones. Explain the problem and fix.

Hardware performance varies with interface condition, so it could explain the timing difference.
Use equivalent hardware across conditions or otherwise design the study to separate the two effects.

Lecture 06 | PDF pp. 28-32

12. Compare between-subjects and within-subjects designs.

Between-subjects uses separate participant groups in each condition, avoiding cross-condition order effects.
Within-subjects uses the same people in all conditions, reducing individual variation but requiring order control.

Lecture 06 | PDF pp. 35-49

13. Explain matched-participants design and its limitation.

Pair participants on relevant characteristics and place members of each pair in different conditions.
Matching reduces selected individual differences but cannot guarantee equivalence on every factor.

Lecture 06 | PDF pp. 35, 49

14. Everyone uses A before B and is faster with B. How would you address the order effect?

Practice or fatigue could affect the second condition independently of interface quality.
Counterbalance exposure by assigning some participants to AB and others to BA.

Lecture 06 | PDF pp. 38-46

15. What does a Latin square contribute to a multi-condition study?

It places each condition once in each sequence position across the planned orders.
This balances position effects with fewer orders than testing every permutation; it does not remove every confound.

Lecture 06 | PDF p. 47

16. Why use a field study to investigate appropriation?

Appropriation concerns how people integrate and adapt technology within their real practices and culture.
Observe natural use and ask contextual questions to reveal adaptations that lab tasks may miss.

Lecture 06 | PDF pp. 51-53

ACTIVE RECALL | SHORT ANSWERS

Two-line models 17-24

Recall before reading the answer. Keep both essential points; equivalent accurate wording can count.

17. Explain why interviews and observation are not restricted to one data type.

Open comments and descriptions provide qualitative evidence about meaning and experience.
Closed responses and counted or timed actions can provide quantitative evidence from the same methods.

Lecture 06 | PDF pp. 57-59

18. Distinguish nominal, ordinal, interval and ratio scales.

Nominal labels categories; ordinal orders them without guaranteed equal gaps.
Interval has equal differences and a conventional zero; ratio adds a meaningful true zero.

Lecture 06 | PDF pp. 62-66

19. Why are recorded age brackets ordinal even though exact age can be ratio data?

Age brackets retain an order but replace exact ages with categories.
The recorded categories do not preserve exact differences or meaningful ratios between individuals.

Lecture 06 | PDF pp. 64, 66, 69

20. For times 2, 2, 3, 4 and 19 seconds, give the mean, median and mode.

Mean = 30 / 5 = 6 seconds; median = the middle value, 3 seconds.
Mode = 2 seconds; the unusually long time raises the mean more than the median.

Lecture 06 | PDF pp. 60-61

21. The nine-person example shows lower times for A than B. State a careful conclusion.

This sample appears to perform the task more efficiently with A than with B.
The descriptive result alone does not establish statistical significance or superiority for all users.

Lecture 06 | PDF p. 71

22. Define content analysis and inter-rater reliability.

Content analysis systematically assigns excerpts to defined or emergent categories and examines patterns or counts.
Inter-rater reliability concerns how consistently different coders assign the same material to categories.

Lecture 06 | PDF pp. 74-77

23. What distinguishes a critical incident from a general impression?

A critical incident is a specific, pivotal event illustrating success or failure in the activity.
Report the observable event and consequence, such as erroneous barcode entry causing a billing dispute.

Lecture 06 | PDF pp. 78-79

24. Distinguish open, axial and selective coding in grounded theory.

Open coding develops categories from data; axial coding relates categories and subcategories.
Selective coding integrates them around a central category while collection and analysis continue to inform each other.

Lecture 06 | PDF pp. 80-91

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 1-8

Choose one best answer. Explain the concept before checking the answer key.

1. A team evaluates a prototype to find problems for its next iteration. What is the purpose?

A. Summative evaluation
B. Random assignment
C. Selective coding
D. Formative evaluation

2. Which activity is analytical evaluation rather than an empirical user study?

A. Observing participants perform tasks
B. Measuring user completion times
C. Inspecting an interface against heuristics without recruiting users
D. Interviewing users during a field trial

3. A researcher teaches every step before measuring first-use learnability. What is the main problem?

A. Training always increases sample representativeness
B. The guidance changes what is being measured
C. Learnability cannot be studied with tasks
D. Guidance makes observations qualitative only

4. Which task prompt avoids giving away the interaction path?

A. Find and enrol in the specified course
B. Click Search, open the first result, then press Enrol
C. Press the green button to finish
D. Use the menu labelled Course Search

5. Which measure most directly addresses memorability?

A. Number of installed features
B. Immediate liking of the colour palette
C. Network capacity during download
D. Task performance after a specified period away from the product

6. Which pairing is correct?

A. Task completion time: utility; successful completion: colour contrast
B. Satisfaction rating: objective speed; errors: feature count
C. Successful task completion: effectiveness; completion time: efficiency
D. Number of features: memorability; age: task success

7. Before recording a participant's face and voice, what does this lecture require?

A. Only permission from the software developer
B. Written, signed consent for the recordings
C. Consent after publication of the video
D. No additional consideration because the test is academic

8. Participants smile and say the interface is easy, but many cannot complete the task. What should be reported?

A. Both the positive attitudes and the observed task failures
B. Only the smiles because they prove effectiveness
C. Only the time of successful users, without mentioning failures
D. A conclusion that satisfaction guarantees task success

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 9-16

Choose one best answer. Explain the concept before checking the answer key.

9. For a positively worded SUS item, a response of 4 contributes how many points before scaling?

A. 4
B. 1
C. 10
D. 3

10. For a negatively worded SUS item, a response of 2 contributes how many points before scaling?

A. 2
B. 0
C. 3
D. 5

11. A participant's ten converted SUS item scores sum to 30. What is the overall SUS score?

A. 30
B. 75
C. 12
D. 100

12. Which interpretation of SUS is sound?

A. A score of 75 is a scaled usability score, not 75% task success
B. A score of 75 proves 75% of tasks succeeded
C. A score of 68 is the 68th percentile by definition
D. A score of 100 proves there are no usability problems

13. In a comparison of old and new enrolment interfaces, which is the independent variable?

A. The measured completion time
B. The number of errors observed
C. The participant's satisfaction response
D. Which interface is used

14. In the same comparison, what is a dependent variable?

A. The assigned interface condition
B. The planned task instructions
C. Time taken to complete the defined enrolment task
D. The decision to counterbalance

15. All old-interface users receive a slow computer, while all new-interface users receive a fast one. Why is this problematic?

A. The task is necessarily qualitative
B. Computer performance is confounded with interface condition
C. Both conditions now have identical hardware
D. The independent variable has disappeared

16. Participants are randomly allocated to use either A or B, and each uses only one. Which design is this?

A. Between-subjects design
B. Within-subjects design
C. Grounded theory
D. Content analysis

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 17-24

Choose one best answer. Explain the concept before checking the answer key.

17. Every participant uses both interfaces A and B. Which design is this?

A. Between-subjects design
B. Nominal classification
C. A field study by definition
D. Within-subjects design

18. Participants are paired by relevant expertise, then one member uses A and the other B. Which design is this?

A. Unmatched within-subjects design
B. A Latin square by definition
C. Matched-participants design
D. Selective coding

19. Everyone uses A first and B second; B is faster. What alternative explanation needs attention?

A. The second condition must have less utility
B. Practice or order effects may benefit the second condition
C. Randomization has eliminated all alternatives
D. A larger monitor was necessarily used

20. Which procedure counterbalances two within-subject conditions?

A. Some participants use A then B; others use B then A
B. Everyone uses A then B
C. One participant uses only A and all others only B
D. The researcher hides all timing data

21. What does a Latin square primarily balance?

A. Every possible individual difference
B. The number of questionnaire words
C. All possible interaction effects without assumptions
D. Each condition's position in the sequence

22. What is a major benefit of within-subjects comparison?

A. Order effects cannot occur
B. No tasks need to be defined
C. Each person serves as their own comparison, reducing between-person variation
D. Every population is represented automatically

23. What distinguishes research experiments from typical product-focused usability testing in this lecture?

A. Product tests never plan their procedure
B. Research emphasizes generalizable knowledge, replicability and statistical validation
C. Research experiments do not need participants
D. Product tests are forbidden from collecting numbers

24. Which study best reveals how people adapt a nutrition app during real grocery shopping?

A. A field study in the real shopping context
B. Only a heuristic inspection in an office
C. A lab task with all interruptions removed
D. Only counting the number of screens

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 25-32

Choose one best answer. Explain the concept before checking the answer key.

25. What is appropriation in this lecture?

A. Randomly assigning users to conditions
B. Multiplying a SUS total by 2.5
C. Calculating a sample median
D. Users integrating and adapting technology to their needs and practices

26. Which statement about acquisition method and data type is most accurate?

A. Every interview response is qualitative
B. Every questionnaire answer is a ratio measurement
C. Interviews and observations can produce both qualitative and quantitative evidence
D. Video can never be used to count errors

27. Numbers 1, 2 and 3 identify unordered user groups. What scale is this?

A. Ordinal
B. Nominal
C. Interval
D. Ratio

28. What scale describes age brackets such as 21–30 and 31–40?

A. Ordinal
B. Nominal with no ordering
C. Ratio merely because ages are mentioned
D. Interval with guaranteed equal category spacing

29. Which is an interval-scale example in the lecture?

A. Task duration in seconds
B. Unordered country labels
C. A first/second/third preference ranking
D. Temperature measured in degrees Celsius

30. Why can 40 seconds be described as twice 20 seconds for task duration?

A. All ordered categories allow ratios
B. Seconds are nominal identifiers
C. Duration is a ratio measure with a meaningful zero
D. Duration has no meaningful differences

31. Which statement about a single agree/disagree Likert item is defensible?

A. The response numbers automatically prove equal intervals
B. The responses are ordered, but equal gaps are not guaranteed
C. Strong agreement is twice agreement
D. The lowest category is a true absence of attitude

32. Task times are 2, 2, 3, 4 and 19 seconds. Which is correct?

A. Mean = 6; median = 3; mode = 2
B. Mean = 3; median = 6; mode = 19
C. Mean = 2; median = 4; mode = 3
D. Mean = 19; median = 2; mode = 6

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 33-40

Choose one best answer. Explain the concept before checking the answer key.

33. Which choice describes variation rather than central tendency?

A. Mean and median
B. Mode and mean
C. Median and mode
D. Range and standard deviation

34. In a boxplot, what does the line inside the box usually represent?

A. The sample mean by definition
B. The maximum
C. The median
D. The number of participants

35. System A has a lower median task time than B for nine tested people. Which conclusion is appropriately limited?

A. A is statistically superior for every population
B. The sample appears faster with A; this alone does not establish population-wide superiority
C. The two systems have identical error rates
D. A must also have higher satisfaction

36. Of 49 respondents, 30 agree and 12 strongly agree that dropping a course is easy. What proportion is positive?

A. 42/49, approximately 85.7%
B. 30/49, approximately 61.2%
C. 12/49, approximately 24.5%
D. 42/100, exactly 42%

37. Which description best fits content analysis?

A. Treat every quotation as a representative population average
B. Discard categories before reading the data
C. Use only completion times and never words
D. Systematically code excerpts into categories and examine their occurrence

38. Two researchers code the same recordings. What does inter-rater reliability address?

A. Whether the interface is error-free
B. Whether their theory is certainly true
C. How consistently their category assignments agree
D. Whether their sample size is automatically sufficient

39. A cashier mistypes a barcode after scanning fails, causing a billing dispute. What is the critical incident?

A. A general claim that all customers dislike the store
B. The failed scan followed by erroneous manual entry
C. The researchers' opinion that the interface looks dated
D. The total number of supermarket products

40. Which sequence best represents the grounded-theory coding approach shown?

A. Open: identify categories; axial: relate them; selective: integrate around a core category
B. Open: calculate SUS; axial: rank users; selective: choose a colour
C. Open: prove a theory; axial: ignore contradictory data; selective: stop analysis
D. Open: assign interfaces; axial: time tasks; selective: randomize participants

CHECK YOUR REASONING

MCQ answers 1-20

1. D - Formative evaluation informs changes during design; summative evaluation assesses the finished product.

Lecture 06 · PDF p. 12

2. C - Analytical methods inspect or predict interaction; empirical evaluation collects evidence from users.

Lecture 06 · PDF pp. 5, 10–11

3. B - Step-by-step coaching can conceal first-use difficulties; any necessary training should match the study goal and be reported.

Lecture 06 · PDF p. 15

4. A - A goal-based task lets researchers observe whether users can discover the necessary actions.

Lecture 06 · PDF pp. 13–15

5. D - The lecture links performance after time away with remembering how to use the system.

Lecture 06 · PDF p. 16

6. C - Effectiveness concerns achieving the goal; efficiency concerns the resources used to do so.

Lecture 06 · PDF pp. 13, 16

7. B - The lecture explicitly requires written-signed consent before these recordings.

Lecture 06 · PDF pp. 14, 17

8. A - Experience reports and performance evidence answer different questions and can disagree.

Lecture 06 · PDF pp. 17–18, 22–23

9. D - Positive items contribute response minus 1: 4 - 1 = 3.

Lecture 06 · PDF p. 19

10. C - Negative items contribute 5 minus response: 5 - 2 = 3.

Lecture 06 · PDF p. 19

11. B - Multiply the converted total by 2.5: 30 × 2.5 = 75 on the 0–100 scale.

Lecture 06 · PDF pp. 19–21

12. A - SUS summarizes questionnaire responses; performance and percentile interpretation require separate evidence or a reference distribution.

Lecture 06 · PDF pp. 19–23

13. D - The researcher varies interface condition and measures resulting performance or experience.

Lecture 06 · PDF pp. 30–32

14. C - The dependent variable is the measured response, rather than the manipulated factor.

Lecture 06 · PDF pp. 30–32

15. B - Hardware changes systematically with interface, so either factor may explain a timing difference.

Lecture 06 · PDF pp. 28–32

16. A - Separate participant groups experience separate conditions.

Lecture 06 · PDF pp. 35–37

17. D - The same participants appear in both conditions, allowing within-person comparison.

Lecture 06 · PDF pp. 35, 38–42

18. C - Matching reduces differences on selected characteristics, but cannot ensure equivalence on everything.

Lecture 06 · PDF pp. 35, 49

19. B - Participants may improve through prior practice or change through fatigue, regardless of interface quality.

Lecture 06 · PDF pp. 38–46, 49

20. A - AB/BA distributes the order of interface exposure instead of tying each interface to one position.

Lecture 06 · PDF pp. 43–46

CHECK YOUR REASONING

MCQ answers 21-40

21. D - Each condition appears once in each ordinal position; balanced variants can additionally address immediate carryover.

Lecture 06 · PDF p. 47

22. C - Comparing the same people across conditions reduces between-person differences, but order effects still require attention.

Lecture 06 · PDF pp. 35–49

23. B - The lecture contrasts their typical purposes and rigor, not an absolute ban on overlap in methods.

Lecture 06 · PDF pp. 24, 50

24. A - Field studies examine natural activity, context and appropriation of technology.

Lecture 06 · PDF pp. 51–53

25. D - Appropriation concerns how technology is adopted and fitted to users' needs, desires and culture.

Lecture 06 · PDF p. 52

26. C - An open comment is qualitative; a closed response or counted event can be quantitative, regardless of collection method.

Lecture 06 · PDF pp. 57–59

27. B - Numbers used as category labels do not establish order or equal distances.

Lecture 06 · PDF pp. 62–63

28. A - The recorded categories are ordered; they do not retain each person's exact age or a true-zero quantitative scale.

Lecture 06 · PDF pp. 64, 69

29. D - Celsius differences are meaningful, but its zero is conventional, so 20°C is not twice the temperature of 10°C.

Lecture 06 · PDF pp. 65–66

30. C - Ratio scales support meaningful ratios because zero represents absence of the measured quantity.

Lecture 06 · PDF p. 66

31. B - A single ordered response item is ordinarily ordinal; numeric labels alone do not establish equal intervals.

Lecture 06 · PDF pp. 64–68, 72

32. A - The sum is 30, so mean is 6; the middle ordered value is 3; 2 occurs most often.

Lecture 06 · PDF pp. 60–61

33. D - Variation describes spread; mean, median and mode describe typical or central values.

Lecture 06 · PDF p. 60

34. C - The box spans the central quartiles, with a line marking the median; whisker conventions should be stated.

Lecture 06 · PDF pp. 70–71

35. B - The lecture explicitly limits the example conclusion to the particular sample; time alone does not prove every other quality.

Lecture 06 · PDF p. 71

36. A - Combine agree and strongly agree, then divide by the actual respondent total: 42/49.

Lecture 06 · PDF p. 72

37. D - Categories can be predefined or emerge from data; coding provides a systematic route from excerpts to findings.

Lecture 06 · PDF pp. 74–77

38. C - Rater agreement concerns coding consistency; agreement alone does not prove that the interpretation is valid.

Lecture 06 · PDF p. 74

39. B - Critical incidents focus on specific pivotal facts rather than broad impressions of dissatisfaction.

Lecture 06 · PDF pp. 78–79

40. A - Grounded theory iterates between data and analysis; codes, relationships and an integrating account develop from evidence.

Lecture 06 · PDF pp. 80–91